

Table 1: System Parameters and Simulation Configuration

Parameter	Value	Description
Number of Users (N_u)	200	Total users in cell
Number of Files (N_f)	2000	Content library size
Cache Size (C)	200	Cache storage capacity
Cache Ratio	10.0%	Percentage of catalog cached
Zipf Exponent (α)	1.0	Content popularity skew
Transmit Power (P_t)	1.0 W	Base station Tx power
Noise Power (N_0)	1.00e-09 W	AWGN power
Cell Radius (R)	500.0 m	Cell coverage radius
Path Loss Exponent (γ)	3.5	Urban/suburban model
Target Rate (R_{target})	0.5 bps/Hz	QoS requirement
Power Coeff. Weak (p_w)	0.8	Power to weak user
Power Coeff. Strong (p_s)	0.2	Power to strong user
SIC Imperfection (ζ)	0.05	Residual interference
SNR Range	-10 to 28 dB	Signal-to-noise sweep
Monte Carlo Runs	50	Statistical replications

Table 2: Channel Model Specifications

Component	Model/Description
Large-Scale Fading	Path loss: $PL(d) = d^{-\gamma}$, $\gamma = 3.5$
Small-Scale Fading	Rayleigh: $ h ^2 \sim \text{Exp}(1)$
User Distribution	Uniform in circle, radius 500.0 m
Channel Gain	$g = PL(d) \times h ^2$
Weak User Gain	$g_w = \min\{g_1, g_2\}$
Strong User Gain	$g_s = \max\{g_1, g_2\}$
SINR (Weak User)	$\gamma_w = \frac{P_t p_w g_w}{P_t p_s g_w + N_0}$
SINR (Strong User)	$\gamma_s = \frac{P_t p_s g_s}{N_0 + \zeta P_t p_w g_s}$
Achievable Rate	$R = \log_2(1 + \gamma)$ bps/Hz
BER (BPSK)	$P_e = \frac{1}{2} \text{erfc}(\sqrt{\gamma})$

Table 3: NOMA Transmission Operations

Operation	Description
User Pairing	Pair weak user (low g_w) with strong user (high g_s) Strategy: Extreme pairing (weakest $\leftrightarrow$ strongest)
Power Allocation	Grid search over $p_w \in [0, 1]$, $p_s = 1 - p_w$ Maximize users satisfying $R \geq R_{target}$ Weak user gets higher power: $p_w = 0.8$
Weak User Decoding	Direct decoding of own signal Treats strong user as interference SINR: $\gamma_w = \frac{P_t p_w g_w}{P_t p_s g_w + N_0}$
Strong User SIC	1. Decode weak user signal first 2. Subtract (imperfect, factor ζ) 3. Decode own signal SINR: $\gamma_s = \frac{P_t p_s g_s}{N_0 + \zeta P_t p_w g_s}$

Table 4: Cache-Aided NOMA Operations

Phase	Operation
Placement Phase	
Content Popularity	Model as Zipf distribution: $p_i \propto i^{-\alpha}$
Cache Strategy	Store top- C most popular files
Cache Size	$C = 200$ files (10.0% of catalog)
Delivery Phase	
Request Generation	Users request files according to Zipf
Cache Hit	File served locally at rate R_{cache} No wireless transmission needed Outage probability = 0, BER = 0
Cache Miss	File transmitted via NOMA Subject to channel conditions Outage probability > 0, BER > 0
Performance Metrics	
Sum-Rate	$R_{sum} = R_w + R_s$ (per-request basis)
Outage Prob.	$P_{out} = P(R < R_{target})$
BER	$P_e = \frac{1}{2}\text{erfc}(\sqrt{\gamma})$ for BPSK